PROPOSAL FOR OPEN SOURCE LAB

Submitted by

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OPEN SOURCE LAB AT MIDDLE SCHOOL

This proposal is for a cutting-edge computer lab that would serve as a showcase for the School District's commitment to using the best that technology has to offer to meet the educational needs of its students. Lessons learned from this project could lead to the construction of more labs with the ultimate goal of providing 1-to-1 computing for its students.

Open Source Software

This proposal utilizes Open Source software, which is secure and reliable, highly customizable to optimize the learning environment, and requires no licensing fees. Accordingly, the district could see substantial cost savings while increasing students' access to computers. This has been the experience of the following that have adopted Open Source software:

- State of Indiana
- City of Atlanta, Georgia
- San Diego County, California
- Bainbridge Island, Washington
- Riverdale School District, Oregon

The Middle School Open Source Lab outlined in this proposal is based in part on the Open Source Case Study for Riverdale High School (see http://www.k12ltsp.org/rhs_casestudy.html). The Riverdale High School created two Open Source labs, each with 35 computers. Their lab supports the Macintosh, Windows, and Linux operating systems. Their lab required an investment of approximately $100,000.

This proposal is for a single lab for 25 students. Only one operating system, Linux, will be used. The largest single cost of the proposed program is the infrastructure costs to upgrade the electrical and data wiring in the classroom to support the computer systems. This has been estimated to be $25,000. Costs to purchase new hardware for the lab are estimate at $15,000, as shown on the attached "Hardware List." As stated on the top of the Hardware List, hardware costs may be substantially reduced by negotiating with vendors, using existing hardware, and soliciting donations from companies and individuals.

The Open Source paradigm is the wave of the future. Contributions to human knowledge and entertainment now come from around the world. Consider the impact of Wikipedia, Facebook, and YouTube on our society today and the potential for the future. The Web 2.0 paradigm is based on the notion that people, including students, should not merely be passive acceptors of technology and media, but that they should have an active role in creating it. Open Source software is part of this trend.

Not only would this proposal lay the groundwork to put the School District on the cutting edge in the adoption of educational technology, but the proposed lab would position the school as a feeder program for the AAA technology school of the future.

Curriculum

The curriculum enabled by the lab proposed here is designed to engage students' curiosity and make them excited about learning through technology. This curriculum advances the five performance standards for Applied Learning.

In particular, the lab is planned to divide a classroom of 25 students into groups of five. Most of the learning follows a constructivist model where the students discover knowledge while accomplishing group and individual activities. The group activities will involve a component of self-evaluation and of the work of others in the group. Also, each class will culminate with group reports on what was accomplished that day. Finally, each student will be expected to develop his or her web portfolio to keep track of challenges met and projects completed.

The first performance standard for applied learning is for problem solving. Almost every project in the curriculum requires students to solve some kind of problem, individually or as part of a group. The second standard, communication tools and techniques, is advanced by students' web portfolios, images and animations, audio recordings, and oral presentations to the class. The third standard, information tools and techniques, is met by the web portfolios, images and animations, and audio recordings, as well as computer programming and use of spreadsheets. The fourth standard, learning and self-management tools and techniques, is advanced by the self evaluations that accompany most web-portfolio reports. Finally, the fifth standard, tools and techniques for working with others, is served by the group-project paradigm, including especially the intra-group written critiques on projects.